

Introduction to Causal Inference

Javier Larrosa

UPC

larrosa@cs.upc.edu

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Motivation

In many situations we want to make an educated guess about what will happen if we take certain action. In other words, we want to predict the **consequences** of actions.

Example (Ibuprofen)

I want to know if my headache will go away if I **take** an Ibuprofen

Example (Policy Making)

We may want to know about the evolution of global warming under different policies. (is it worth **subsidizing** electric cars?)

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- Only Experimental Research seems to be suitable for proving Causality
- In this lecture we will try to understand more deeply this idea

Association - Dependence

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We say that there is **association** between two variables (or that they are **dependent** of each other) if, when we **observe** them, we see that they are somehow interconnected

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Observation

The association between two variables is bi-directional

Association vs Causation

Association does not imply Causation

In the well-known statement **correlation does not imply causation** the word correlation means **association** or **dependence**

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Causation implies Association

Causation implies association, but not the other way around

Example

- In humans there is association between **height** and **reading ability**, but not causation.
- In humans there is association and causation between **diet** and **obesity**. In fact, the causation goes from diet to obesity.

Observation

Causality is directional

Back to Human Irrationality

- When we **observe** the world, it is relatively easy to identify associations
- Humans tend to (wrongly) see causation way too often (that is, jumping from association to causation)
- It is important (and very difficult in general) to know when the causation is real and its strength

Back to Human Irrationality

- When we **observe** the world, it is relatively easy to identify associations
- Humans tend to (wrongly) see causation way too often (that is, jumping from association to causation)
- It is important (and very difficult in general) to know when the causation is real and its strength
- Classic Statistics deals with associations (i.e. conditional probability, correlation, linear regression, etc)
- Data Science and Machine Learning use as input observed data and capture associations (not causation)
- To know more: [The Book of Why](#)

Definition (Confounder)

A variable X is a confounder of T and Y if it is dependent with both T and Y , and **produces association** between them.

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Age is a confounder for the association between **height** and **reading ability**

Observation

Confounders are the main source of confusion in our cause-effect understanding of the world

Statistics has always been interested in measuring associations. It can also be used to measure causality.

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Definition

- $P(Y = y)$ probability of random variable Y taking value y
- $P(Y = y|X = x) = \frac{P(Y=y, X=x)}{P(X=x)}$ probability of random variable Y taking value y **conditioned to** X takes value x

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Example:

- Probability of being healthy is Spain $P(H = 1) = 0.7$
- Probability of being healthy conditioned to being stressed $P(H = 1| S = 1) = 0.5$

Definition (Independence)

Variable Y is independent of variable X if for all $y \in Y$ and for all $x \in X$,

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Observation

Variable independence is bi-directional. If X is independent of Y , then Y is independent of X

Law of Total Probability

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Example:

$$\begin{aligned} P(H = 1) &= \\ &= P(H = 1 | S = 1)P(S = 1) + \\ &\quad + P(H = 1 | S = 0)P(S = 0) = \\ &= 0.5 \times 0.01 + 0.702 \times 0.99 = 0.7 \end{aligned}$$

Definition (Expectation)

The Expectation of random variable X is,

$$E[X] = \sum_{x \in X} P(X = x) \cdot x$$

Property

If X is a boolean random variable (i.e, $X \in \{0, 1\}$, Bernouilli), its expectation is,

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Example:

The expected value of random variable "being healthy" is

$$E[H] = 0.3 \times 0 + 0.7 \times 1 = 0.7$$

Measuring Association

Measuring Association

Consider a simple case in which we consider two boolean random variables T and Y . A measure of association is,

$$E[Y|T = 1] - E[Y|T = 0]$$

because it measures how different is variable Y in the cases where $T = 1$ and the cases where $T = 0$

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Example:

- The association between being healthy and being stressed is $0.5 - 0.702$. It is high because the stressed population and the non-stressed population are significantly different with respect to health
- The association between being healthy and being blonde is $0.7 - 0.7$. It is zero because the blonde population and the non-blond population are identical with respect to health

Measuring Association: Example with confounder

Confounders produce association.

- Consider the following data about somebody's life where S means that he woke up with his **shoes on** and H means that he woke up with **headache**,

S	H	numb. of cases	$P(S,H)$
0	0	793	0.793
0	1	97	0.097
1	0	27	0.027
1	1	83	0.083

- Association:

$$E[H|S = 1] - E[H|S = 0] = \frac{83}{27+83} - \frac{97}{793+97} = 0.75 - 0.1 = 0.65$$

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- In this example we know that there is no causality at all
- Association is produced by the obvious **confounder** D meaning that he was drunk the previous night!!

Measuring Association: Example with not so obvious confounder

- Consider the following data about a set of homes, where B means that there are **many books** in the house and S means that children brought up in that house ended up having a **university degree**,

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- An obvious **confounder** is a third variable E meaning that the parents have a high socio-economic level
- But there could also be some **causality** from books to academic success
- I would be great if we could measure how much of the association is indeed causal**

Potential Outcomes Framework

We are ready now to explore a simple formal approach to Causal Inference

Definition (Potential Outcomes Framework)

Let T represent a possible **intervention** (aka **treatment**, **exposure**, **action**) and Y represent the possible *observable consequences* of the intervention (aka **outcome**)

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- Population: people
- Treatment T : hours of weekly exercise along 3 months
- Outcome Y : cholesterol change

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Example

- Population: strawberry farms
- Treatment T : amount of fertilizer
- Outcome Y : production in Kg.

Hypothetical Interventions

You can only apply this framework if you can, at least hypothetically, manipulate T

Definition (Immutable)

Variables that cannot be manipulated are called immutable

Example

- Race, sex, age are immutable
- Weight can be manipulated, but it is tricky because it cannot be done directly (drugs, exercise, diet, surgery,...)
- Smoking can be manipulated (hypothetically, because it is unethical)
- A medical treatment can be manipulated (drug or placebo)

Hypothetical Interventions

If in our causal question the treatment variable T cannot be hypothetically manipulated, sometimes we can rephrase it into one with a more clear manipulable variable

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Example

Suppose that we want to understand the effect of race on job finding. We cannot manipulate the race of people.

Hypothetical Interventions

If in our causal question the treatment variable T cannot be hypothetically manipulated, sometimes we can rephrase it into one with a more clear manipulable variable

Example

Suppose that we want to understand the effect of race on job finding. We cannot manipulate the race of people.

We can study, instead, how the name in the resume affects the probability of being appointed for an interview.

Potential Outcomes: Notation

In the following, for simplicity, we will assume that our causal question is about the effect of certain drug on a certain patient.

- we will assume boolean variables (treatment $T \in \{0, 1\}$ and outcome $Y \in \{0, 1\}$)
- $T = 1$ means that the patient is treated with the drug
- $T = 0$ means that the patient is not treated
- $Y = 1$ means that the patient recovers
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- $Y = 1$ means that the patient recovers
- $Y = 0$ means that the patient does not recover
- Y^t represents the outcome that **will be** observed if $T = t$ (**potential outcome**)

Factuals and Counterfactuals

When we consider two possible alternative actions and we take one of them (for instance $T = 1$)

- **Factual:** consequence of the action that we took (Y^1)
- **Counterfactual:** consequence of the action that we did not take (Y^0)

Example

- I have a headache and I can either take an ibuprofene ($T = 1$) or not ($T = 0$)
- Before I decide my action, the possible consequences of the possible actions are Y^t (note that there are 4 possibilities)
- At this point Y^t are **potential outcomes**
- Let's suppose that I decide to take the ibuprofen ($T = 1$)
- Y^1 is the **factual**
- What would have had happend if I did not take the ibuprofen (Y^0) is the **counterfactual**

Example

Let's think about what happened in the previous story.

- I had a headache. I took an ibuprofen (that is $T = 1$) and the headache is gone (that is $Y^1 = 1$)
- Can I claim: **therefore the ibuprofen worked?**

Example

Let's think about what happened in the previous story.

- I had a headache. I took an ibuprofen (that is $T = 1$) and the headache is gone (that is $Y^1 = 1$)
- Can I claim: **therefore the ibuprofen worked?** NOOO!
- I need to compare Y^1 with Y^0
- but this is impossible! We never know the counterfactual
- This is the **Fundamental problem of causal inference**

Fundamental problem of causal inference

Definition (Fundamental problem of causal inference)

It is impossible to know counterfactuals

Overcoming the Fundamental Problem

- Since it is impossible to know if $Y^0 \neq Y^1$ for a particular individual, we will work with populations and Statistics

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- Since it is impossible to know if $Y^0 \neq Y^1$ for a particular individual, we will work with populations and Statistics
- We would like to compare the expected value of Y^0 (that is, $E[Y^0]$) with the expected value of Y^1 (that is, $E[Y^1]$)
 - $E[Y^0]$ prob. of recovery in the population if **nobody** took the drug
 - $E[Y^1]$ prob. of recovery in the population if **everybody** took the drug

Average Treatment Effect (ATE)

$E[Y^1] - E[Y^0]$ is a measure of the causal effect of the treatment.

Overcoming the Fundamental Problem

Average Treatment Effect (ATE)

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Without Confounders

If there are no confounders it can be shown that,

$$E[Y^t] = E[Y|T = t]$$

therefore,

$$ATE = E[Y|T = 1] - E[Y|T = 0]$$

In words: the expected value of the outcome if all the population is treated as $T = t$ is equal to the expected value of the outcome in those cases in which $T = t$

Without confounders, causality=association

Overcoming the Fundamental Problem

With confounders (Stratification or adjustment formula)

- Let T and Y be a treatment and the outcome of interest, respectively.
- Let X be confounder variable (assume just 1 for simplicity).

Under some technical conditions then,

$$E[Y^t] = \sum_{x \in X} E(Y|T = t, X = x)P(X = x)$$

therefore,

$$\begin{aligned} ATE &= E[Y^1] - E[Y^0] \\ &= \sum_{x \in X} E(Y|T = 1, X = x)P(X = x) - \\ &\quad - \sum_{x \in X} E(Y|T = 0, X = x)P(X = x) \end{aligned}$$

Example: Shoes On and Headache, revisited

- Recall the data where S means that he woke up with his shoes on and H means that he woke up with headache. We want to know if sleeping wearing shoes causes headache.

S	H	numb. of cases	$P(S,H)$
0	0	793	0.793
0	1	97	0.097
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If there were no confounders:

- Causality=Association:

$$E[H|S = 1] - E[H|S = 0] = \frac{83}{27+83} - \frac{97}{793+97} = 0.75 - 0.1 = 0.65$$

- The causality would be very high

Example: Shoes On and Headache, revisited

But being drunk D is an obvious confounder. Therefore, we need to include it in the data

S	H	D	numb. of cases
0	0	0	792
0	0	1	1
0	1	0	88
0	1	1	9
1	0	0	18
1	0	1	9
1	1	0	2
1	1	1	81

- $E[H^1] = P(H = 1|S = 1, D = 0)P(D = 0) + P(H = 1|S = 1, D = 1)P(D = 1) = 0.18$
- $E[H^0] = P(H = 1|S = 0, D = 0)P(D = 0) + P(H = 1|S = 0, D = 1)P(D = 1) = 0.18$
- $ATE = 0.18 - 0.18 = 0$ There is no Causality

Measuring Association: Example with not so obvious confounder

- Recall the example about a set of homes, where B means that there are **many books** in the house and S means that children brought up in that house ended up having a **university degree**,

B	S	P(B,S)
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- If we believe that there is no confounder: Association=causality
$$E[S|B = 1] - E[S|B = 0] = \frac{0.07}{0.24} - \frac{0.1}{0.76} = 0.29 - 0.13 = 0.16$$

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 $E[S|B = 1] - E[S|B = 0] = \frac{0.07}{0.24} - \frac{0.1}{0.76} = 0.29 - 0.13 = 0.16$
- If we believe that the Socio-economic level of the families is (the only) confounder we need to incorporate this data

Measuring Association: Example with not so obvious confounder

B	S	L	P(B,S,L)
0	0	0	0.6
0	0	1	0.06
0	1	0	0.04
0	1	1	0.06
1	0	0	0.15
1	0	1	0.02
1	1	0	0.01
1	1	1	0.06

- $E[S^1] = P(S = 1|B = 1, L = 0)P(L = 0) + P(S = 1|B = 1, L = 1)P(L = 1) = 0.2$
- $E[S^0] = P(S = 1|B = 0, L = 0)P(L = 0) + P(S = 1|B = 0, L = 1)P(L = 1) = 0.15$
- $ATE = 0.2 - 0.15 = 0.05$ Most of the association was due to the confounder, but there is some causality

Causal Graphs Framework

- Now that we understand the basics of the Potential Outcome Framework we are going to see how it extends to richer scenarios
- We will use a framework called Causal Graphs.

Definition (Causal Graphs)

Causal graphs are directed acyclic graphs (DAGs) where:

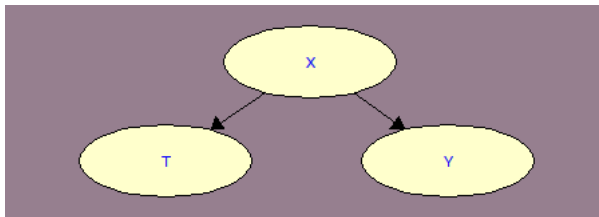
- Nodes correspond to variables of interest
- Arcs correspond to possible direct causality

Definition (Causal Path)

In a Causal Graph, a Causal Path is a path $X_1, X_2, \dots, X_k$ such that there is an arc $X_i \rightarrow X_{i+1}$ for each step of the path

Let's see the simplest 3-nodes cases.

Confounder



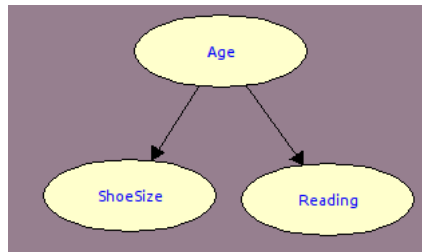
- There is no causal path from T to Y
- There is association from T to Y and from Y to T .
- The association goes through the **confounder** X

Note that association also goes through paths but it may go against the arrows!!

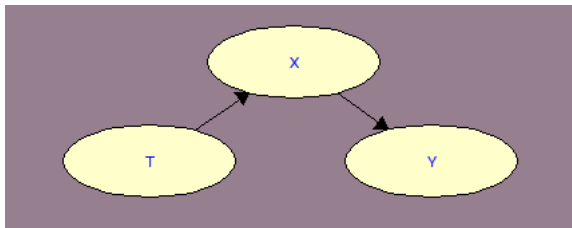
Example

Example

Let A denote the age of a person, S denote his shoe size, R his reading ability.



Mediator



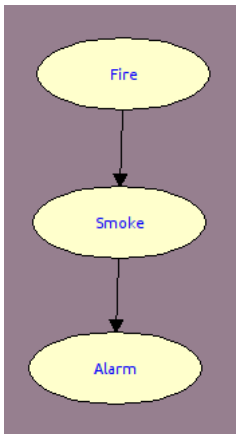
- There is a causal path from T to Y
- The causation is indirect (through the **mediator** X)
- There is association from T to Y and from Y to T

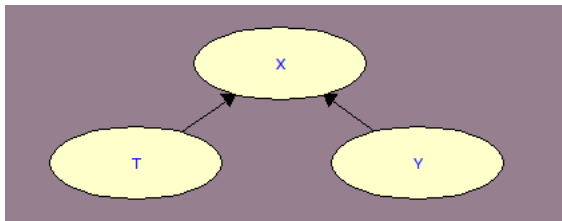
Again, association may flow through paths against arrows!!

Example

Example

Let F denote fire, S smoke, A denote alarm.





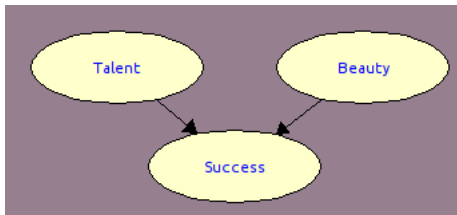
- There is no causation from T to Y
- There is no association from T to Y , nor from Y to T .
- Variable X is called a **collider**

Association does not flow through colliders!!!

Example

Example

Let S denote how successful is an artist, T denotes his, B denote his beauty.



Causal Inference in a 3-node Causal Graph

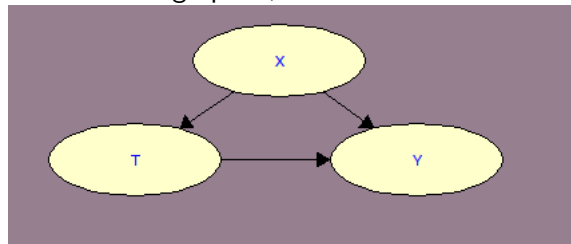
Consider a table of frequencies,

T	X	Y	numb. of cases
...	...	...	...
...			

Where we want to assess the causality from T to Y

Causal Inference in a 3-node Causal Graph: Confounder

If the Causal graph is,

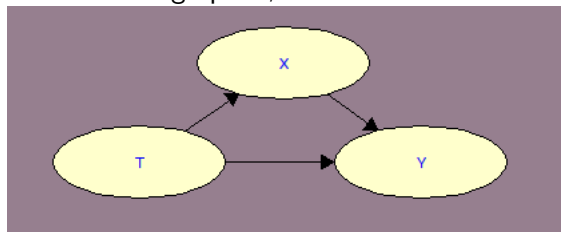


X is a **confounder** between T and Y

- The adjustment formula **has to be used**
- The effect of adjusting X is to discount the association that is not causal

Causal Inference in a 3-node Causal Graph: Mediator

If the Causal graph is,

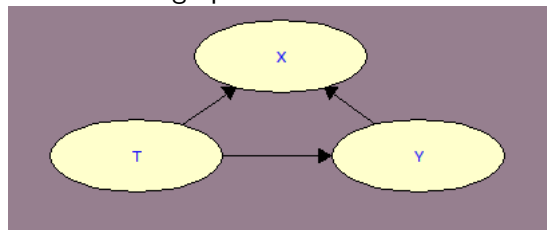


X is a **mediator** between T and Y

- The adjustment formula **does not have to be** used
- If used, it would discount the association going through the mediator

Causal Inference in a 3-node Causal Graph: Collider

If the Causal graph is

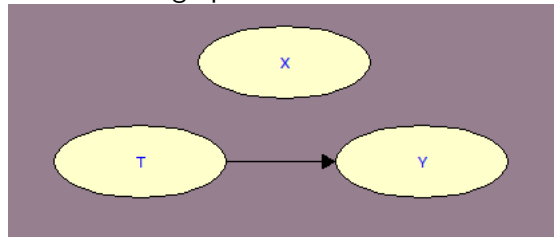


X is a **collider** between T and Y

- The adjustment formula **does not have to be** used
- If used, it would create spurious association going through the collider

Causal Inference in a 3-node Causal Graph: Independent

If the Causal graph is



X is unrelated with T and Y

- The adjustment formula **does not have to be** used
- But, if used, it would be useless (up to adding some noise in the estimations)

Example

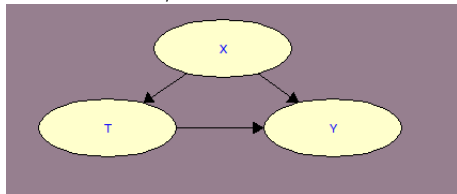
Suppose that we want to assess whether studying (T) has any effect in the obtained marks (Y). For simplicity we will assume boolean variables ($T = 0$ means did not study and $Y = 0$ means did not pass the course) Suppose that we get the following data from TMIRI students

T	Y	frequency
0	0	35
0	1	45
1	0	10
1	1	10

- $E[Y] = \frac{55}{100} = 0.55$
- $E[Y|T = 0] = \frac{45}{80} = 0.56$
- $E[Y|T = 1] = \frac{10}{20} = 0.5$
- The association is small (variables seem to be almost independent)
- If there are no confounders, it seems that studying is slightly counterproductive (what??)

Example Cont.

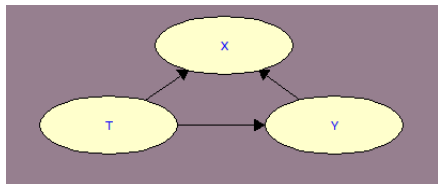
But there may be confounders. A very reasonable one is the intelligence of a student, since it may affect his/her studying habits. Thus, we enlarge our model as,



which means that we have to enlarge our data and adjust our computations,

Example Cont.

- In the previous slide we assumed that intelligence was an objective measure (from an IQ test or something like that)
- However, let's suppose that intelligence was obtained asking the parents opinion
- Probably, the opinion of parents is influenced by their son's studying habits and passing records
- Which means that the causal graph is more likely to be,



which means that we do not have to enlarge our data and adjust our computations: studying seems to be slightly counterproductive.

Simpson's paradox

Example (Kidney Stones)

A hospital treats kidney stones with two different treatments, A and B. The (observational) data from the hospital is

Rec. Rate	A	B
	$273/350 = 78\%$	$284/350 = 81\%$

Which treatment is more effective? B

Simpson's paradox

Example (Kidney Stones)

We also have information about the size of the stones (Large or Small). If we take this data into account,

Rec. Rate	A	B
Total	$273/350 = 78\%$	$284/350 = 81\%$
Small	$81/87 = 93\%$	$234/270 = 87\%$
Large	$192/263 = 73\%$	$50/80 = 63\%$

Simpson's paradox

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Rec. Rate	A	B
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Large	$192/263 = 73\%$	$50/80 = 63\%$

Which treatment is more effective?

- For small stones A
- For large stones A

This situation is known as **Simpson's paradox**

Simpson's paradox

To deal with the paradox we need to know the story behind the data.

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Scenario 1:

- Treatment A requires surgery (riskier) and Large stones are more challenging.

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- Therefore, Size is a confounder
 - Size affects treatment
 - Size affects chance of recovery

Simpson's paradox

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Scenario 1:

- Treatment A requires surgery (riskier) and Large stones are more challenging.
- It turns out that doctors likely assign treatment A to Large stones.
- Therefore, Size is a confounder
 - Size affects treatment
 - Size affects chance of recovery
- Therefore, we need to adjust for Size

$$E[rec^{treat}] = \sum_{size} E(rec|treat, size)P(size)$$

- $E(rec^A) = 0.83$ and $E(rec^B) = 0.75$. Therefore treatment A is better